District-Level Child-Marriage Prevalence in India: Prediction, High-Prevalence Classification and Correlates using Classical Machine-Learning Models

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Abstract. Marriage before the legal age of 18 harms girls' health, schooling and autonomy, and its prevalence still ranges from 0% to about 58% across Indian districts. This study asks how well classical machine-learning (ML) models taught in an undergraduate syllabus can estimate district-level child-marriage prevalence from other district health, education and living-condition indicators, and which indicators matter most. We use the NFHS-5 (2019-21) district fact-sheet data for 706 districts of 36 States/UTs. The target is the percentage of women aged 20-24 married before 18. After cleaning suppressed cells, median imputation, winsorisation and standardisation, nine regressors and eight classifiers were tuned with five-fold cross-validation and compared with a mean predictor and a state-mean predictor. A support-vector regressor gave the lowest test RMSE (5.18 percentage points, $R^2 = 0.813$), 57% below the mean predictor and 38% below the state-mean predictor, while regularised linear models were within 4%. Flagging districts above the all-India figure of 23.3% worked best with logistic regression (F1 = 0.862, ROC-AUC = 0.941). Holding out entire states gave a lower but still useful R^2 of about 0.70. The strongest correlates were teenage motherhood, women's schooling and literacy, and sanitation; when teenage motherhood, which partly measures the same phenomenon, was removed, R^2 fell to 0.76. Simple, interpretable models are adequate for district-level screening, but they describe association, not cause, and are weakest in the states where prevalence is highest.

Keywords: child marriage; NFHS-5; India; district-level analysis; regression; classification; support vector machine; lasso; state-grouped cross-validation; SDG 5; SDG 4

1 Introduction

Child marriage, defined here as marriage before the legal minimum age of 18 for women, is linked to early childbearing, lower schooling, poorer nutrition and reduced decision-making power. India has made large gains: the share of women aged 20-24 married before 18 fell from 47.4% in NFHS-3 (2005-06) to 26.8% in NFHS-4 and 23.3% in NFHS-5 (2019-21) [1]. These national averages hide wide gaps. In our data, district prevalence ranges from 0% to 57.6%, and the NFHS-5 sample is designed to give estimates down to the district level, which is where many programmes under the Prohibition of Child Marriage Act and related schemes are actually run.

A district-level view raises a practical ML question. Programme officers have many indicators for every district but little time; can a model that uses those indicators tell them *how high* prevalence is likely to be, and *which* districts are above the national average? Because the target is itself a survey estimate, a good model is also a way to ask which other indicators move together with child marriage. This links the work to SDG 5 (Gender Equality, target 5.3: eliminate child, early and forced marriage) and SDG 4 (Quality Education). Our contributions are: (i) a documented cleaning protocol for the NFHS-5 district file, including its suppressed and low-sample cells; (ii) a comparison of nine regression and eight classification models against two simple baselines; (iii) an evaluation on *unseen states* using state-grouped cross-validation, which is closer to real use than a random split; and (iv) a leakage check showing how much performance comes from indicators that are partly consequences of child marriage.

Section 2 reviews related work, Section 3 describes the data, Sections 4-5 the methods and set-up, Section 6 the results and Section 7 the discussion, followed by the conclusion.

2 Related Work

Raj, Dehingia and co-authors [2] applied lasso, ridge and neural-network models to more than 4,000 variables from NFHS-4 women's records to identify correlates of child marriage and note that survey data of this kind support association, not causation. Dietrich et al. [3] predicted where child marriage is most likely to occur in South Asia from regional inputs such as droughts, floods, population growth and night-light data, and found that gradient boosting identified most true cases. Singh, Shekhar and Shri [4] studied age at first marriage across five NFHS rounds with survival models and found that region, education, caste, religion, wealth and media exposure were significantly associated with it. These studies work with individual records or remote-sensing inputs; we instead use the published district-level indicators that a district officer can actually see, and we test the models on states they have not seen. For this we follow the argument that random cross-validation is optimistic when observations are spatially or hierarchically clustered [5]. Interpreting district-level correlations as individual-level effects would be an ecological fallacy [6], so we report associations only. All models were implemented with scikit-learn [7]; the algorithms follow [8], with lasso from [9].

3 Dataset and Preprocessing

3.1 Source and description

The data file *datafile.csv* contains the NFHS-5 (2019-21) district fact-sheet indicators published by the International Institute for Population Sciences (IIPS) and the Ministry of Health and Family Welfare [1], as distributed on Kaggle [10]. It has 706 rows (districts) and 109 columns: district name, State/UT, three sample-size columns and 104 percentage or count indicators on households, education, marriage and fertility, family planning, maternal and child health, nutrition, anaemia, blood pressure, blood sugar, cancer screening, and tobacco and alcohol use. Our target is the column *Women age 20-24 years married before age 18 years (%)*. It is complete for all 706 districts, with mean 20.4%, median 18.2%, standard deviation 12.4 percentage points (pp), and a right-skewed distribution (Fig. 2). Sixteen rows share a district name with another row (for example Aurangabad in Bihar and in Maharashtra), so districts are identified by the pair (district, State/UT). Two State labels needed repair ('Maharastra' and a leading space in ' Lakshadweep').

3.2 Suppressed cells and missing values

The file stores several indicators as text. A value in brackets, such as (29.5), means the estimate rests on only 25-49 unweighted cases; there are 5,068 such cells and we keep their values. An asterisk (*) means fewer than 25 cases and the value is suppressed; there are 4,125 such cells and we treat them as missing. Six indicators were missing in more than 40% of districts (Fig. 1; between 60% and 91%) and were dropped, as were the three sample-size columns, which measure survey effort rather than a district characteristic. The remaining 97 candidate predictors have 1.4% missing cells overall (in 23 columns); these were filled with the training-set median inside the model pipeline so that test data never influence the imputation.

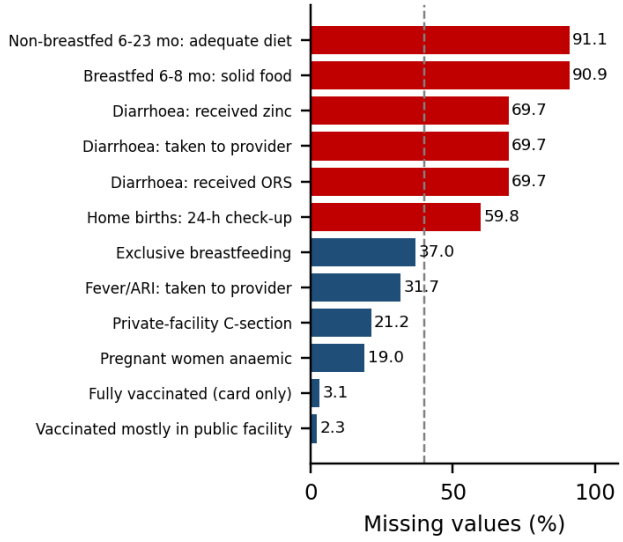


Fig. 1. Percentage of districts with missing (suppressed) values for the twelve most affected indicators (red: above the 40% threshold; dropped).

3.3 Label, outliers and scaling

For the classification task a district is labelled *high prevalence* (c = 1) if its prevalence exceeds the NFHS-5 all-India value of 23.3% [1]; 36.3% of districts (256 of 706) fall in this class, so the classes are moderately balanced. Percentages are already bounded between 0 and 100, so there are no impossible values, but a few indicators have extreme districts. Because extreme districts are real, not errors, we winsorised every feature at its 1st and 99th training percentiles rather than deleting rows, and then standardised, $z = (x - \mu)/\sigma$, with μ and σ estimated on the training fold only. Figure 2 shows the target distribution and its strong clustering by State: the State alone explains 57% of the variance of the district values (in-sample), with Bihar (41.1% mean), Tripura (39.7%) and West Bengal (39.4%) highest and Kerala (5.5%) and Jammu & Kashmir (4.8%) lowest.

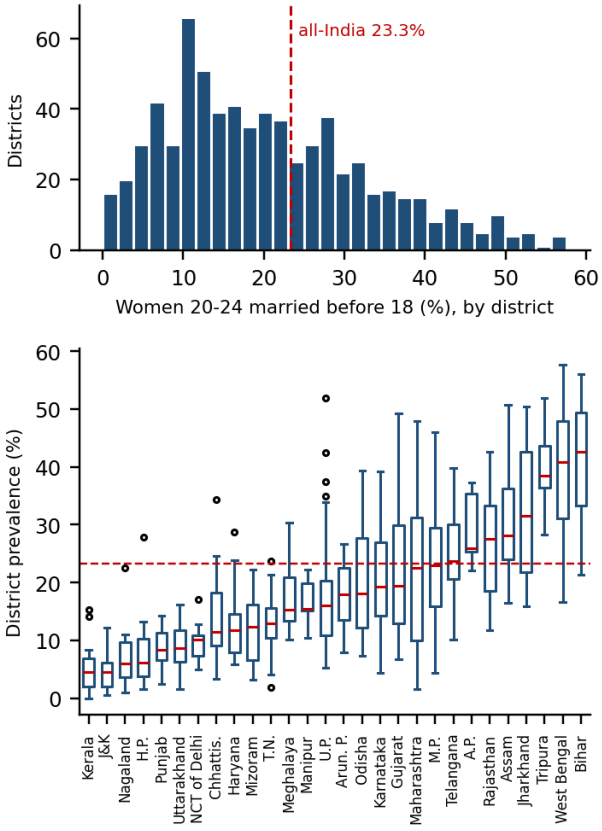


Fig. 2. Top: distribution of district prevalence (dashed line: all-India 23.3%). Bottom: box plots by State/UT for States with at least eight districts, ordered by median.

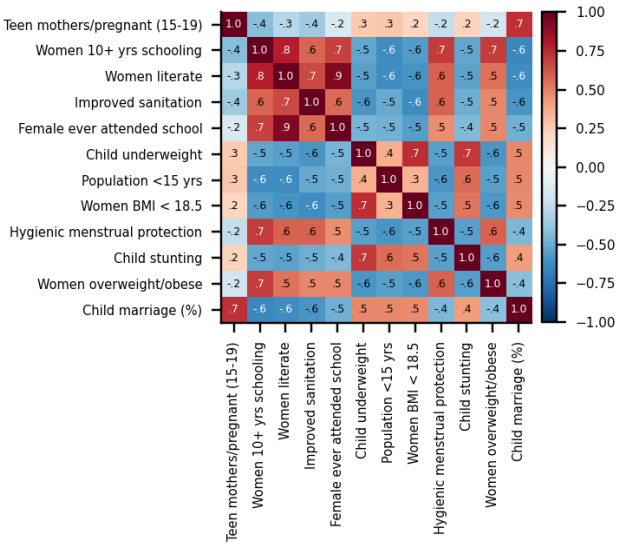


Fig. 3. Pearson correlation matrix of the eleven indicators most correlated with the target, and the target (training districts only).

3.4 Correlation and PCA

On the training districts, the indicators most positively correlated with child-marriage prevalence were teenage motherhood or pregnancy among women aged 15-19 (r = 0.73), child underweight (r = 0.48), the share of the population under 15 (r = 0.48) and women's low BMI (r = 0.48). The most negative were women with 10 or more years of schooling (r = -0.65), women's literacy (r = -0.62) and improved sanitation (r = -0.58) (Fig. 3). Many predictors are strongly inter-correlated (for example, literacy and 10+ years of schooling, r = 0.77; child stunting and underweight, r = 0.70), which motivates regularisation. The tuned lasso set 44 of 97 coefficients exactly to zero, an embedded form of feature selection. PCA on the 97 standardised features (Fig. 4) needed 36 components for 90% and 49 for 95% of the variance; the first component explained 21.4%. The slow growth of the curve shows that the indicators carry many partly independent signals.

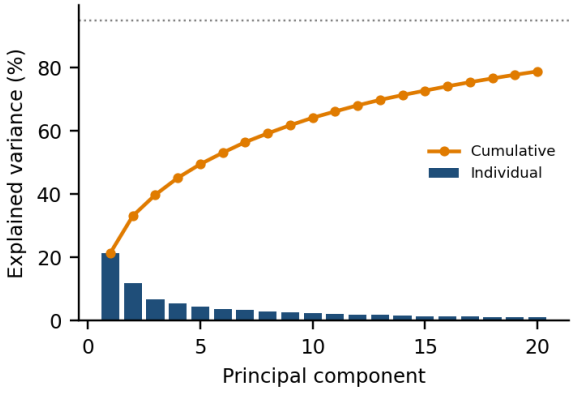


Fig. 4. Individual and cumulative explained variance of the first 20 principal components (training data).

4 Methodology

Fig. 5 summarises the end-to-end workflow. All steps after the split are wrapped in a single scikit-learn pipeline that is re-fitted on the training part of every fold.

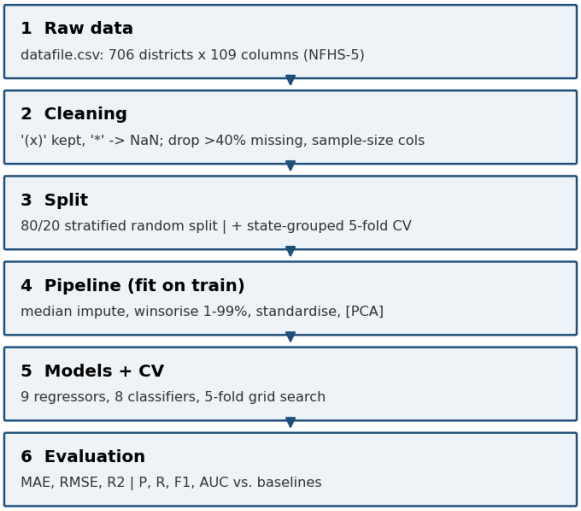


Fig. 5. Methodology pipeline used in this study (square brackets: optional step).

Let **x** be the standardised indicator vector of a district and y its child-marriage prevalence in %. Two simple baselines are used. The **mean predictor** outputs the training mean, and the **state-mean predictor** outputs the mean of the training districts of the same State (the global mean for a State with no training district):

$$\hat{y}_d = \text{mean}(\, y_j : j \text{ in training, State}(j) = \text{State}(d)\,) \quad (1)$$

4.1 Regression models

Multiple linear regression fits $\hat{y} = \mathbf{w}^{\mathsf{T}}\mathbf{x} + b$ by minimising the mean squared error

$$J(\mathbf{w}) = (1/n)\, \Sigma_i\, (y_i - \hat{y}_i)^2,\ \mathbf{w} \leftarrow \mathbf{w} - \eta\, \partial J/\partial \mathbf{w} \quad (2)$$

where the second expression is the gradient-descent update (scikit-learn uses the closed-form solution, which reaches the same minimum). **Ridge** and **lasso** add L2 and L1 penalties to control over-fitting and multicollinearity:

$$J_{ridge} = J + \lambda\, \Sigma_j\, w_j^2,\ J_{lasso} = J + \lambda\, \Sigma_j\, |w_j| \quad (3)$$

Polynomial regression expands the features to all degree-2 terms followed by ridge. A **regression tree** recursively splits the feature space to minimise within-node squared error. **k-NN regression** averages the targets of the k nearest training districts under Euclidean distance. **Support-vector regression** (SVR) ignores errors smaller than ε and penalises larger ones linearly:

$$\min \tfrac{1}{2}\|\boldsymbol{w}\|^2 + C\,\Sigma_i(\xi_i + \xi_i^*),\ |y_i - f(\boldsymbol{x}_i)| \le \varepsilon + \xi_i \quad (4)$$

with the RBF kernel $K(\mathbf{x}, \mathbf{x}') = \exp(-\gamma\|\mathbf{x} - \mathbf{x}'\|^2)$. The **multilayer perceptron** (MLP) has two hidden layers (64 and 32 ReLU units), is trained by back-propagation with Adam, an L2 penalty α and early stopping. A **random forest** (200 trees) averages decorrelated trees [11]; it is included as an extra reference model.

4.2 Classification models

The label is c = 1 if the prevalence is above 23.3%. **Logistic regression** models

$$P(c = 1 \mid \boldsymbol{x}) = \sigma(\boldsymbol{w}^T\boldsymbol{x} + b),\ \sigma(z) = 1 / (1 + e^{-z}) \quad (5)$$

and is trained by minimising regularised cross-entropy (inverse strength C). **Gaussian Naive Bayes** assumes conditionally independent normal features: $P(c \mid \mathbf{x}) \propto P(c)\,\Pi_j\, N(x_j;\, \mu_{jc},\, \sigma^2_{jc})$. The **decision tree** minimises Gini impurity $1 - \Sigma_k\, p_k^2$; **k-NN** takes a majority vote of the k nearest districts; the **perceptron** updates $\mathbf{w} \leftarrow \mathbf{w} + \eta(c_i - \hat{c}_i)\mathbf{x}_i$ on each mistake; the soft-margin **SVM** with RBF kernel solves the classification analogue of (4); the **MLP classifier** mirrors the regressor with a logistic output; and a **random forest** is again included for reference. Class weighting ('balanced') was treated as a hyper-parameter where the model supports it.

4.3 Evaluation metrics

Regression: $MAE = (1/n)\Sigma|y - \hat{y}|$, $RMSE = \sqrt{[(1/n)\Sigma(y - \hat{y})^2]}$ and $R^2 = 1 - \Sigma(y - \hat{y})^2 / \Sigma(y - \bar{y})^2$; errors are in percentage points (pp). Classification: accuracy, precision = TP/(TP+FP), recall = TP/(TP+FN), F1 = 2PR/(P+R) and the area under the ROC curve (AUC). For a screening use, a missed high-prevalence district (false negative) is costlier than a false alarm, so recall and F1 are emphasised over accuracy. Confidence intervals for the test metrics use 2,000 bootstrap resamples of the test districts.

5 Experimental Setup

Main split. The 706 districts were split at random 80/20, stratified by quartile of the target: 564 training and 142 test districts (random seed 42). The high-prevalence rate is 36.2% in training and 36.6% in the test set.

Cross-validation and tuning. Hyper-parameters were chosen by grid search with five-fold cross-validation on the training districts (shuffled K-fold for regression, stratified K-fold for classification), optimising RMSE for regression and F1 for classification. Imputation, winsorisation and scaling sit inside the pipeline and are re-fitted on every training fold, which prevents leakage. Table 2 lists the grids and selected values.

Unseen states. Districts in the same State share policies, culture and survey design, so a random split lets the model learn State-specific levels from neighbouring districts. To measure how well a model works in a State it has not seen, we also ran *state-grouped five-fold cross-validation* on all districts (GroupKFold: no State appears in both the training and test part of a fold), using the hyper-parameters already chosen on the main split.

Leakage check. Some indicators are partly *consequences* of child marriage, notably teenage motherhood or pregnancy (women aged 15-19) and the share of third-or-higher-order births. We therefore repeat the regression with these two removed (feature set B, 95 indicators) and with only 16 structural indicators on education, sanitation, electricity, clean fuel, insurance, registration, demography and menstrual hygiene (set C), and with State indicators added (set A+State).

Environment. Python 3, scikit-learn 1.8, pandas, NumPy, SciPy and Matplotlib; random seed 42 for all stochastic models. The full code (run_experiments.py) regenerates every number and figure in this paper.

Table 2. Hyper-parameter grids and values selected by five-fold CV (cw = class weight).

Model	Grid searched	Selected (reg. / clf.)
Ridge / Lasso	α in {0.1…1000} / {0.01…1}	alpha=30 / alpha=0.1
Poly-2 + Ridge	α in {1000, 3000}	alpha=1000
Tree	depth {3,4,6,8}; min leaf {5,10,20}	depth=6, leaf=20 / depth=6, leaf=10
k-NN	k {5,10,15,25}; uniform/distance	k=5, w=distance / k=10
SVR / SVM (RBF)	C {1,10,100} (clf. {0.1,1,10}); ε {1,3}; γ {scale, 0.003, 0.01}	C=100, eps=3, gamma=0.003 / C=10, gamma=0.003
MLP	α {0.001, 0.1, 1, 10}	alpha=1 / alpha=0.001
Random forest	max features {0.3, sqrt}; min leaf {1,3}	mf=0.3, leaf=1 / mf=0.3, leaf=1
Logistic reg.	C {0.01…10}; cw	C=0.1, cw=balanced
Naive Bayes	var. smoothing {1e-9…1}	var_smoothing=0.1
Perceptron	α {1e-4…1}; cw	alpha=0.0001, cw=balanced

6 Results

6.1 District prevalence regression

Table 3. Regression on the held-out test districts (percentage points, pp). CV-RMSE is the mean over the five training folds. Best test RMSE in bold.

Model	MAE	RMSE	R^2	CV-RMSE
Mean predictor	9.72	11.98	0.000	-
State-mean predictor	6.35	8.36	0.513	-
Linear reg. (OLS)	4.65	5.75	0.770	6.48
Ridge reg.	4.41	5.41	0.796	6.16
Lasso reg.	4.40	5.38	0.799	6.03
Polynomial (deg 2) + Ridge	5.19	6.86	0.672	6.76
Decision tree	5.77	7.44	0.615	7.83
k-NN regressor	5.17	6.79	0.679	6.91
SVR (RBF)	**4.23**	**5.18**	**0.813**	**5.71**
MLP (64-32)	4.69	5.80	0.766	6.49
Random forest	4.60	5.51	0.788	6.19

Every learned model beats both baselines (Table 3). The SVR achieved the lowest RMSE (5.18 pp, 95% bootstrap CI 4.64-5.73) and highest R^2 (0.813), 57% below the mean predictor (RMSE 11.98) and 38% below the state-mean predictor (8.36). The gain over the state-mean predictor is 2.1-4.2 pp in RMSE (95% CI), so the indicators carry information well beyond the State. Lasso (5.38) and ridge (5.41) were within 4% and 4% of the best model, and the random forest (5.51) within 6%; plain OLS (5.75) was 11% worse, showing that regularisation helps with 97 correlated indicators and 564 training districts. The high-variance models did worst: polynomial ridge (6.86), k-NN (6.79) and the single tree (7.44). With about 100 dimensions and few hundred points, distance-based and deep-tree methods have too little data to fill the space. The test errors are somewhat lower than the CV errors (for the SVR 5.18 vs 5.71), within the sampling noise of a 142-district test set.

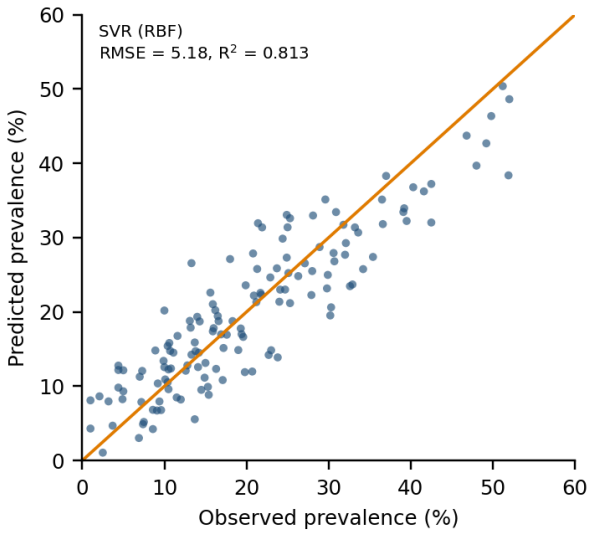


Fig. 6. Observed versus predicted prevalence for the best regressor on the test set; the orange line is perfect prediction.

Fig. 6 shows close agreement over most of the range. The model slightly over-predicts the cleanest districts and under-predicts the very highest-prevalence districts, a typical regression-to-the-mean effect that matters because those are exactly the districts a programme most needs to find.

6.2 High-prevalence classification

Table 4. High-prevalence (above 23.3%) classification on the test districts. Best F1 in bold.

Model	Acc.	Prec.	Recall	F1	AUC
Majority class	0.634	0.000	0.000	0.000	-
State-mean rule	0.761	0.705	0.596	0.646	0.852
Logistic regression	**0.894**	**0.825**	**0.904**	**0.862**	**0.941**
Gauss. Naive Bayes	0.768	0.634	0.865	0.732	0.872
Decision tree	0.866	0.867	0.750	0.804	0.861
k-NN	0.866	0.837	0.788	0.812	0.916
Perceptron	0.824	0.721	0.846	0.779	0.922
SVM (RBF)	0.894	0.863	0.846	0.854	0.951
MLP (64-32)	0.852	0.816	0.769	0.792	0.940
Random forest	0.887	0.846	0.846	0.846	0.938

Logistic regression had the best F1 (0.862, 95% CI 0.79-0.92; recall 0.904), followed by the RBF-SVM (0.854) and the random forest (0.846). The SVM had the highest AUC (0.951), and thresholding the SVR's predicted prevalence at 23.3% also works well (F1 = 0.835, AUC = 0.954), so one regression model can serve both purposes. All main models clearly beat the state-mean rule (F1 0.646), whose recall is only 0.596. Gaussian Naive Bayes was weakest (F1 0.732) because its independence assumption is badly violated by the correlated indicators, and the MLP and single tree were also below the leaders. For logistic regression the confusion matrix was TN = 80, FP = 10, FN = 5, TP = 47: it found 47 of the 52 high-prevalence test districts and missed 5 (about one in 10), with 10 false alarms. The 95% intervals are wide because the test set is small, so differences of 0.01-0.03 in F1 between the top models should not be over-interpreted.

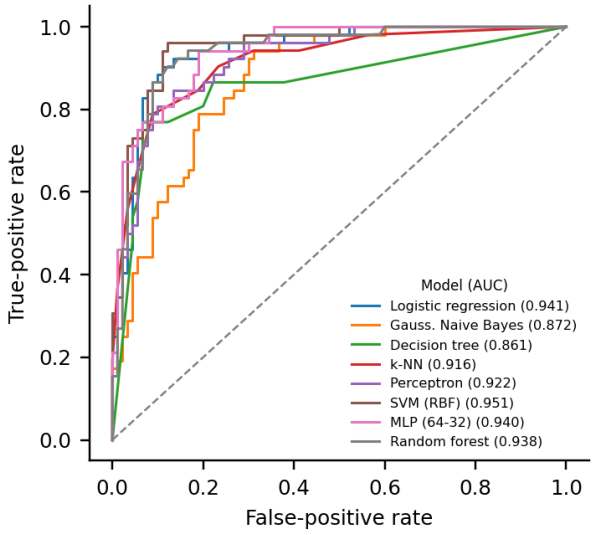


Fig. 7. ROC curves of the eight classifiers on the test set (AUC in brackets).

6.3 Unseen states and robustness

Table 5. Regression on unseen States. Top: state-grouped five-fold CV over all 706 districts (four best models and the random forest). Bottom: test RMSE of three models by feature set (A: all indicators; B and C: see text), and the SVR's unseen-state RMSE (not run for A + State).

Model (state-grouped CV)	MAE	RMSE	R^2
Mean predictor	10.23	12.50	-0.009
Lasso reg.	5.41	6.79	0.702
SVR (RBF)	5.39	6.82	0.699
Ridge reg.	5.51	6.94	0.689
Linear reg. (OLS)	5.89	7.39	0.647
Random forest	5.95	7.47	0.640

Feature set	n	Ridge	SVR	RF	SVR R^2	SVR unseen-state
A: all indicators	97	5.41	5.18	5.51	0.813	6.82
B: A − teen motherhood − birth order	95	6.23	5.89	6.56	0.758	8.05
C: structural indicators only	16	7.93	7.59	7.02	0.599	10.81
A + state indicators	133	5.31	5.16	5.42	0.815	n/a

When whole States are held out, performance drops but stays useful: the best model (Lasso reg.) has RMSE 6.79 and R^2 = 0.702, against RMSE 12.50 for the mean predictor; the SVR gives 6.82 (R^2 = 0.699). The same SVR scored RMSE 5.49 under ordinary random five-fold CV on all districts, so a random split understates the error one should expect in a new State by about 1.3 pp. The flexible models suffer most (random forest 7.47, k-NN 9.57, polynomial ridge 9.30), while the regularised linear models and the SVR are the most stable, which supports using simple models.

Leakage check. Removing teenage motherhood and birth order (set B) raises the SVR's test RMSE from 5.18 to 5.89 (R^2 0.813 → 0.758) and its unseen-state RMSE from 6.82 to 8.05. So a meaningful part of the headline accuracy comes from an indicator that partly measures the same phenomenon. With only the 16 structural indicators (set C) the best model reaches R^2 of about 0.66 (RMSE 7.02), still far better than the mean predictor but clearly weaker. Adding State indicators changed the SVR's RMSE only from 5.18 to 5.16, which shows that the indicators already carry most of the State-level signal.

PCA. Table 6 shows that PCA (95% of the variance, 49 components) helped both models on this data: OLS improved from RMSE 5.75 to 5.20, because projecting onto uncorrelated directions removes the multicollinearity that hurts plain OLS, and k-NN improved from 6.64 to 6.49. Neither beat the regularised models in Table 3.

Table 6. Test RMSE with all features vs PCA (95% variance).

Setting	MAE	RMSE	R^2
OLS, all features	4.65	5.75	0.770
OLS, PCA 95%	4.26	5.20	0.812
k-NN (k=15), all features	5.05	6.64	0.693
k-NN (k=15), PCA 95%	4.94	6.49	0.707

6.4 Which indicators matter?

Three views agree (Fig. 8, last page before the appendix). In the lasso, the largest standardised coefficient is teenage motherhood or pregnancy (+6.1 pp per standard deviation), followed by women's literacy (-2.6) and 10+ years of schooling (-2.2). In the random forest, permuting teenage motherhood raises the test RMSE by 3.3 pp, far more than any other indicator, followed by women's schooling (0.72), literacy (0.63) and improved sanitation (0.22). Without the teenage-motherhood indicator, women's schooling (1.65), literacy (0.75) and sanitation (0.47) become the leading features. Some lasso signs are counter-intuitive (for example, a positive coefficient for institutional births, and a negative one for women's tobacco use); because the indicators are strongly correlated, individual coefficients can flip sign and should be read as part of the whole model, not as separate effects.

7 Discussion

What worked. A modest set of education, sanitation, demographic and health indicators explains most of the district-to-district variation: the best model reaches R^2 of 0.813 on held-out districts and about 0.70 on unseen States. Regularised linear models and the SVR were nearly tied and far more stable than flexible models, and they are interpretable. For a State or district planning office, logistic regression is the natural screening tool: it has the best F1, and its coefficients can be explained.

Where it is weakest. Fig. 9 and Appendix Table A1 show the error of the unseen-state predictions by State. The model beat the mean predictor in 35 of 36 States/UTs (the exception: Andaman & Nicobar Islands). But the average error hides a systematic pattern: when a State is held out, its mean is under-predicted most in West Bengal (-8.0 pp), Rajasthan (-6.0) and Karnataka (-3.3), and over-predicted in Tamil Nadu (+6.5), Jammu & Kashmir (+6.5) and Uttar Pradesh (+5.3). In plain terms, the model tends to pull estimates towards the national average: several of the highest-prevalence States are rated too low and several of the lowest are rated too high. Since the high-prevalence States are where attention is most needed, a screening tool should be tuned for high recall. States with few districts (Table A1) have very noisy error estimates.

Limitations. (i) The data are one cross-section (NFHS-5, 2019-21), so the models cannot show change over time; NFHS-6 will allow a true temporal test. (ii) Only 706 districts are available, so the test set has 142 districts and the confidence intervals are wide. (iii) The district estimates come from a sample survey and carry sampling error themselves, and 4,125 cells were suppressed. (iv) Several indicators, above all teenage motherhood, partly reflect child marriage rather than cause it, which the leakage check quantifies. (v) District-level correlations cannot be read as effects on individual girls (the ecological fallacy [6]); they show where indicators move together, not why. (vi) The hyper-parameter grids are small to keep the study reproducible on a laptop, and survey weights and the sampling uncertainty of each district estimate were not used.

Ethics and SDG impact. The data are aggregate district indicators from a public survey with no personal information. The main ethical risk is misuse: a model that rates a district low could divert attention from it, and a model that rates a district high should not be used to stigmatise communities. A deployed screening tool should favour recall, publish its uncertainty, be audited across States, and be used only to prioritise where programmes, schools and enforcement are strengthened. Reliable district screening supports SDG target 5.3 (eliminating child, early and forced marriage) and SDG 4 (education), since schooling is the most stable correlate we found.

8 Conclusion and Future Work

Using NFHS-5 district indicators for 706 districts and a leakage-aware protocol, we showed that syllabus-level ML models estimate district child-marriage prevalence with $R^2 \approx 0.81$ (RMSE ≈ 5.2 pp) and flag above-average districts with F1 ≈ 0.86, clearly beating a State-average baseline. Performance holds up, at about R^2 0.70, for States the model has never seen, but it falls to about 0.76 once the teenage-motherhood indicator is removed, and the models under-rate the highest-prevalence States. Women's schooling and literacy, sanitation and teenage motherhood are the main correlates. Future work will use NFHS-4 and NFHS-6 for a temporal test of the same models, add district-level socio-economic and administrative data (wealth, caste composition, enforcement), account for the sampling error of each district estimate, and use spatial models that exploit district neighbourhoods.

Declarations

Data availability. The NFHS-5 district fact-sheet data are published by IIPS and the Ministry of Health and Family Welfare [1]; the file used here (datafile.csv, 706 rows, 109 columns) was supplied with the assignment pack and corresponds to the Kaggle dataset 'National Family Health Survey (NFHS-5) 2019-21' [10]. Code and figures: run_experiments.py and build_paper.py (a student places the code on GitHub).

Conflict of interest. The author declares no conflict of interest. **Funding.** None.

Use of AI tools. The analysis code (run_experiments.py), the figures and the first draft of this paper were produced with the help of an AI assistant (Claude, Anthropic) from the assignment pack, the allocation sheet and the supplied CSV file. [Student: edit this paragraph to say exactly which parts you wrote, ran, checked or changed yourself.] All numbers in this paper are generated by the accompanying code, and the author is responsible for the content.

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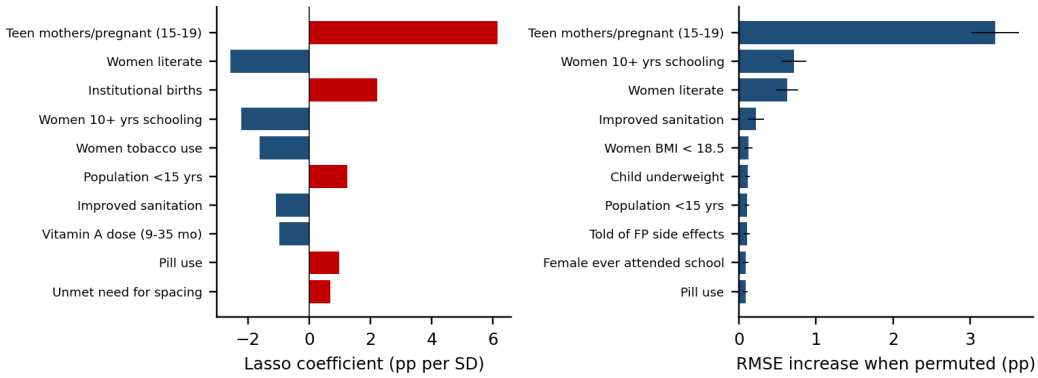


Fig. 8. Left: ten largest lasso coefficients on the standardised scale (red: positive; blue: negative). Right: random-forest permutation importance on the test set (increase in RMSE in pp when one indicator is shuffled; bars show ± 1 SD over 30 repeats).

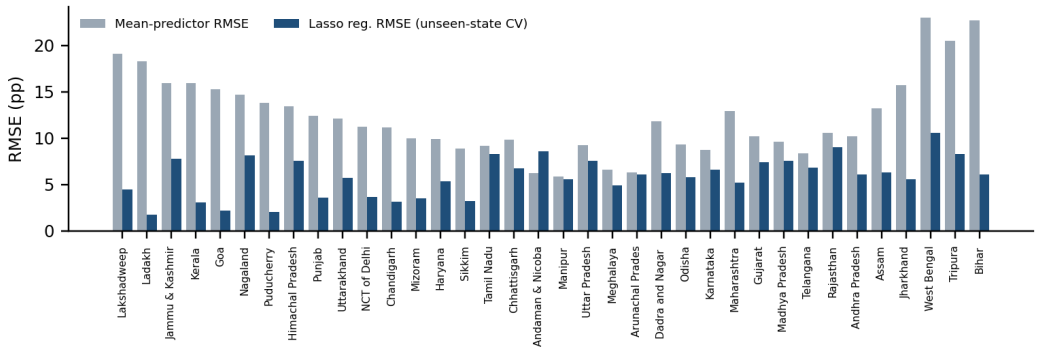


Fig. 9. RMSE by State/UT for the unseen-state predictions of the best grouped model compared with the mean predictor (full table in Appendix A).

Appendix A. Per-State error of the unseen-state predictions

Table A1. State-grouped CV (best model: Lasso reg.). n = number of districts; mean y and mean ŷ = mean observed and predicted prevalence (%); RMSE in pp; the baseline is the training mean of the other States. Bias = mean ŷ − mean y. Rows in bold: States where the model was no better than the baseline.

State/UT	n	Mean y	Mean ŷ	Bias	RMSE model	RMSE baseline
Bihar	38	41.1	38.1	-2.9	6.1	22.7
Tripura	8	39.7	38.5	-1.1	8.3	20.4
West Bengal	19	39.4	31.4	-8.0	10.6	23.0
Jharkhand	24	32.4	31.8	-0.6	5.6	15.7
Assam	33	30.5	27.3	-3.2	6.3	13.2
Andhra Pradesh	13	29.4	31.9	+2.5	6.1	10.2
Rajasthan	33	26.5	20.5	-6.0	9.0	10.6
Telangana	31	24.3	23.2	-1.1	6.8	8.4
Madhya Pradesh	51	23.3	23.9	+0.7	7.5	9.6
Gujarat	33	22.2	20.4	-1.8	7.4	10.2
Maharashtra	36	21.7	22.9	+1.2	5.2	12.9
Karnataka	30	20.5	17.2	-3.3	6.6	8.7
Odisha	30	20.4	20.5	+0.1	5.8	9.3
Dadra and Nagar Haveli & Daman and Diu	3	18.4	13.4	-5.0	6.2	11.8
Arunachal Pradesh	20	18.2	17.9	-0.3	6.1	6.3
Meghalaya	11	17.7	20.1	+2.4	4.9	6.6
Uttar Pradesh	75	16.8	22.2	+5.3	7.5	9.3
Manipur	9	16.5	18.2	+1.7	5.6	5.9
Andaman & Nicobar Islands	**3**	**14.6**	**6.2**	**-8.4**	**8.6**	**6.2**
Chhattisgarh	27	13.7	17.5	+3.8	6.7	9.9
Tamil Nadu	32	12.9	19.4	+6.5	8.3	9.2
Sikkim	4	12.3	13.2	+0.9	3.2	8.9
Haryana	22	12.3	14.0	+1.7	5.3	9.9
Mizoram	8	11.9	10.9	-1.0	3.5	10.0
Chandigarh	1	9.7	6.5	-3.2	3.2	11.2
NCT of Delhi	11	9.6	11.7	+2.1	3.7	11.2
Uttarakhand	13	9.0	11.8	+2.8	5.7	12.1
Punjab	22	8.8	9.2	+0.4	3.6	12.4
Himachal Pradesh	12	8.2	5.2	-3.0	7.6	13.4
Puducherry	4	7.9	6.8	-1.1	2.0	13.8
Nagaland	11	7.4	12.7	+5.3	8.1	14.7
Goa	2	5.7	6.5	+0.8	2.2	15.3
Kerala	14	5.5	5.8	+0.4	3.0	16.0
Jammu & Kashmir	20	4.8	11.3	+6.5	7.8	15.9
Ladakh	2	2.5	4.1	+1.6	1.8	18.3
Lakshadweep	1	1.3	5.8	+4.5	4.5	19.1

Generated by run_experiments.py (per_state_error.csv). States with one to three districts have very uncertain error estimates.